



Elena Tonini

ID: CA06767OV **Passport:** YC4824376 **Nationality:** Italian

Date of birth: 02/06/2000 **Place of birth:** Manerbio, Italy **Gender:** Female

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Home: Viale Italia 4A, 25021 Bagnolo Mella (Italy)

EDUCATION AND TRAINING

PhD in Telecommunications Engineering

University of Brescia [01/11/2024 – Current]

City: Brescia | Country: Italy | Website: <https://www.unibs.it/> | Level in EQF: EQF level 8

Master's Degree in Computer Science and Engineering

University of Brescia [09/2022 – 09/2024]

City: Brescia | Country: Italy | Website: <https://www.unibs.it/> | Final grade: 110 cum Laude | Level in EQF: EQF level 7 | Thesis: Statistical Analysis to Support CSI-Based Sensing Methods

Link: <https://arxiv.org/pdf/2411.03203>

Bachelor's Degree in Computer Science and Engineering

University of Brescia [09/2019 – 10/2022]

City: Brescia | Country: Italy | Website: https://www.unibs.it | Final grade: 106/110 | Level in EQF: EQF level 6 | Thesis: Analysis and Characterization of Wi-Fi Channel State Information

Link: https://ans.unibs.it/assets/documents/thesis/Elena_Tonini_Bach_Thesis.pdf

High School Diploma with specialisation in Classical Studies

Liceo Classico Arnaldo [12/09/2014 – 08/06/2019]

City: Brescia | Country: Italy | Website: <https://liceoarnaldo.edu.it> | Field(s) of study: Arts and humanities | Final grade: 100 cum Laude | Level in EQF: EQF level 4

WORK EXPERIENCE

Technische Universität Berlin - Telecommunication Networks Group (TKN) – Berlin, Germany

Visiting Phd Student

[05/02/2026 – 31/07/2026]

University of Brescia – Brescia, Italy

Website: <https://www.unibs.it/en/university/organisation/departments/information-engineering-dii> | Name of unit or department: Department of Information Engineering - Business or sector: Information and communication

Graduate teaching assistant

[09/2025 – 09/2026]

University of Brescia – Brescia, Italy

Website: <https://www.unibs.it/en/university/organisation/departments/information-engineering-dii> | Name of unit or department: Department of Information Engineering - Business or sector: Information and communication

Undergraduate Teaching Assistant

[11/2023 – 10/2024]

 **IEEE Student Branch of Brixia** – Brescia, Italy

Webmaster and Social Media Manager

[10/2022 – 10/2024]

- Managing the social media channels of the student association (IEEE Student Branch STB1019)
- Engaging with companies interested in collaborating with the association
- Head of the team organising events, meetings, and conferences hosted by IEEE STB1019

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Latin

LISTENING A2 **READING** A2 **WRITING** A2

SPOKEN PRODUCTION A2 **SPOKEN INTERACTION** A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

think analytically / cooperate with colleagues / use microsoft office / use online communication tools / work as a team / use different communication channels / give constructive feedback / use a computer / Latex / work in an international environment / follow work schedule / develop strategy to solve problems / prepare presentation material

PUBLICATIONS

[2026]

Energy-Efficient CSI-Based People Tracking with Spiking Neural Networks

Authors: Alisa Snezkaia, Elena Tonini, Renato Lo Cigno, Giovanni Perin | **Journal Name:** Neuromorphic Physical Layer Signal Processing for Wireless Systems Workshop | **Publisher:** ACM

[2026]

[Inside the Channel: A Stochastic Analysis of the CSI for Wi-Fi Sensing](#)

Authors: Elena Tonini, Renato Lo Cigno, Francesco Gringoli, Marco Cominelli | **Journal Name:** IEEE Journal of the Internet of Things | **Publisher:** IEEE

[2026]

[Delay-Doppler Domain Feature Extraction for Explainable Wi-Fi Activity Recognition via SVM](#)

Authors: Elena Tonini, Renato Lo Cigno, Emanuele Viterbo | **Journal Name:** Mediterranean Artificial Intelligence and Networking Conference (MAIN)

[2026]

[Leveraging Mutual Information in Stochastic CSI Analysis for Wi-Fi Sensing](#)

Authors: Elena Tonini, Renato Lo Cigno | **Journal Name:** 21st Wireless On-demand Network systems and Services Conference

[2025]

[mmWave CSI-based Sensing: a Feasibility Study](#)

Authors: Joana Angjo, Elena Tonini, Falko Dressler, Renato Lo Cigno | **Journal Name:** 31st Annual International Conference on Mobile Computing and Networking | **Volume, Issue and Pages:** pages 1353 - 1355 | **Publisher:** ACM

[2025]

Towards a Quantitative Analysis of CSI for AI/ML Based Sensing

Authors: Elena Tonini, Francesco Gringoli, Renato Lo Cigno, Marco Cominelli | **Journal Name:** 2025 IEEE Wireless Communications and Networking Conference (WCNC) | **Publisher:** IEEE

[2024]

Statistical Analysis to Support CSI-Based Sensing Methods

Authors: Elena Tonini | **Journal Name:** Preprint

[2024]

Can Platoons Form On Their Own?

Authors: Lorenzo Ghiro, Renato Lo Cigno, Elena Tonini, Stefano Fontana, Michele Segata | **Journal Name:** 2024 IEEE Vehicular Networking Conference (VNC) | **Publisher:** IEEE

CONFERENCES AND SEMINARS

[01/07/2026 – 03/07/2026] Palermo, Italy

2026 Mediterranean Artificial Intelligence and Networking Conference Presenter of the paper "Delay-Doppler Domain Feature Extraction for Explainable Wi-Fi Activity Recognition via SVM"

[02/03/2026 – 04/03/2026] Crans Montana (Sierre), Switzerland

Wireless On-demand Network systems and Services Conference 2026 Presenter for the paper "Leveraging Mutual Information in Stochastic CSI Analysis for Wi-Fi Sensing"

Links: <https://opendl.ifip-tc6.org/db/conf/wons/wons2026/1571217424.pdf> | <https://2026.wons-conference.org>

[12/01/2026 – 14/01/2026] Bardonecchia (TO), Italy

Italian Networking Workshop (INW) 2026 Presenter for the paper "Leveraging Mutual Information in Stochastic CSI Analysis for Wi-Fi Sensing"

Link: <https://inw.di.unito.it>

[04/11/2025 – 08/11/2025] Hong Kong, China

The 31st Annual International Conference On Mobile Computing And Networking Presenter for the poster "mmWave CSI-based Sensing: a Feasibility Study"

Link: <https://www.sigmobility.org/mobicom/2025/index.html>

[24/03/2025 – 27/03/2025] Milan, Italy

IEEE Wireless Communications and Networking Conference (WCNC2025) Presenter for the paper "Towards a Quantitative Analysis of CSI for AI/ML Based Sensing".

Link: <https://wcnc2025.ieee-wcnc.org>

[20/01/2025 – 22/01/2025] Moena (TN), Italy

Italian Networking Workshop (INW) 2025 Presenter for the poster "Towards Intelligent Ambient Sensing: Quantitative Analysis of CSI for Room Occupancy Detection"

Link: <https://inw.dei.unipd.it>

VOLUNTEERING

[10/2021 – Current] Brescia

IEEE Student Member Member of the IEEE Student Branch of Brixia (IEEE Student Branch 1019). The student association aims to introduce its members to both the professional and academic worlds through engineering projects and cooperations with companies operating in the fields of computer science and electronics. It is also an organiser of Java and Python programming courses offered to first-year students of the University of Brescia that can be validated as part of their university curriculum.

HONOURS AND AWARDS

[27/02/2025] Istituto Lombardo - Accademia di Scienze e Lettere

"Prof. Emilio Gatti" Graduation Award Elena Tonini has been awarded the prestigious "Prof. Emilio Gatti" Graduation Award by the Istituto Lombardo – Accademia di Scienze e Lettere for the year 2024. The award is dedicated to a Master's thesis investigating new methods of measurement either in a theoretical or empirical framework. Her Master's thesis, titled "Statistical Analysis to Support CSI-Based Sensing Methods", explores Channel State Information (CSI), a set of data describing the evolution of a wireless signal based on interactions with the surrounding environment. The study proposes a statistical model to enhance ambient sensing techniques, with applications in Wi-Fi, 5G, and future 6G networks, contributing to the interpretability of Artificial Intelligence and Machine Learning systems.

Link: <https://www.youtube.com/watch?v=BNA-H3m2gYQ>

HOBBIES AND INTERESTS

Theatre Elena Tonini was a member of a theatre company for 8 years, joining the production of multiple theatre and musical theatre shows.

DRIVING LICENCE

Driving Licence: B